OTOMOTO TORONTO MOTO INC.
99 River Street, Toronto, ON M5A 3P4, Canada · 647-424-1088

Both boxes: what to buy, where, and how to assemble

One bike box under the seat of a Triumph that still has the regular 16-pin socket. One window box at 99 River Street, on the shop Wi-Fi. Build the bike box on the bench first. The window box can be soldered the same day. The second radio stays in the bag until the fuse current is under 50 µA.

Buy links were checked 2 November 2026. Shelf counts move. Prices below are the USD amounts those pages showed that day, before Canadian tax and shipping. This sheet is not a quote and not a radio certification.

Stop if any of these are true. The socket under the seat is the small red 6-pin. The bike is a 2006–2012 triple (K-line on pin 7 of this same plastic). You were about to power the box from pin 16. Current at the fuse is over 50 µA. A 120 Ω resistor is still between CAN H and CAN L. The radio is still at the factory 22 dBm. Customer texts stay on Wix Inbox. This build does not send them.

1. Bike box parts

Search Digi-Key Canada with the keyword in the last column: https://www.digikey.ca/en/products/result?keywords= plus that keyword. Mouser Canada takes the same keyword.

Qty	Buy this	Where	Search
1	Seeed Studio XIAO nRF52840, SKU 102010448. Plain board. Not Sense. Not Plus.	Seeed, in stock at USD 9.99 on 2 Nov 2026. seeedstudio.com/Seeed-XIAO-BLE-nRF52840-p-5201.html	102010448
1	Texas Instruments TPS7A1601, order TPS7A1601DGNT. HVSSOP-8 with the metal pad.	Digi-Key or Mouser. TI page: ti.com/product/TPS7A1601	TPS7A1601DGNT
1	Microchip MCP2515-I/P. DIP-18. On 2 Nov 2026 Digi-Key's restock date was 9 Nov 2026.	LCSC C469036, or Digi-Key when the reel lands. SOIC is allowed if you can solder it.	MCP2515-I/P
1	Texas Instruments SN65HVD230DR, plus an SOIC-8 to DIP adapter.	Digi-Key or Mouser. If the DR reel is gone, SN65HVD231 is the allowed cousin. Do not buy a TJA1050 or MCP2551.	SN65HVD230DR
1	16 MHz crystal, and the two load capacitors its datasheet asks for. An 18 pF crystal usually wants about 22 pF on each leg.	Digi-Key. HC-49 or a smaller can. The firmware number is 16 MHz.	16MHz crystal
1	AO3401A P-channel MOSFET, SOT-23.	Digi-Key or LCSC. Any P-channel that is fully on at −2.5 V and is happy at 3.3 V.	AO3401A
2	2N3904 NPN, TO-92. One switches the divider. One senses the key.	Digi-Key, or any small-signal NPN rated 40 V.	2N3904
1	2N3906 PNP, TO-92. Switches the high side of the divider.	Digi-Key, or any small-signal PNP rated 40 V.	2N3906
1	1N5822 Schottky, 3 A, 40 V. Through-hole.	Digi-Key.	1N5822
1	P6KE18A, 600 W unidirectional TVS. Through-hole.	Digi-Key. If this bike's charging voltage sits above 15 V, use a P6KE20A.	P6KE18A
1	1 A ATM mini blade fuse and a sealed inline holder.	Princess Auto, Canadian Tire, or Digi-Key. Any 1 A automotive mini fuse.	ATM fuse holder
1	3.3 V zener, 500 mW. 1N5226B or BZX55C3V3.	Digi-Key.	1N5226B
1	SAE J1962 male OBD-II pigtail, 16-pin, wires already fitted. Short.	Amazon or an auto-parts counter. Not a 16-pin to red 6-pin adapter. Do not trust the jacket colors.	OBD2 pigtail 16 pin
—	18 AWG red and black for the battery. 22 AWG twisted pair for CAN. A gasketed ABS box and a cable gland. Perfboard, solder, heat-shrink.	The box can be a Hammond 1591 or any small gasketed shell that takes a gland.	—

Resistors and capacitors for the bike box

Qty	Part	Where it goes
1	210 kΩ, 1%	Regulator R1, OUT to FB
2	100 kΩ, 1%	Regulator R2, FB to ground. Divider top, from the 2N3906 collector toward D0.
1	12 kΩ, 1%	Divider bottom, D0 side to ground
5	100 kΩ, 5% is fine	2N3906 base to its emitter. Divider-switch base to ground. Key-sense series. Key-sense base to ground. MOSFET gate pull-up to 3V3.
1	47 kΩ	XIAO D1 to the divider-switch 2N3904 base
2	10 kΩ	2N3906 base to the divider-switch collector. MCP2515 RESET up to the switched 3.3 V rail.
2	10 µF ceramic, X7R, 25 V or more	Regulator IN to ground, and OUT to ground, close to the chip
3	100 nF ceramic	One across the regulator if you want the datasheet's extra cap. One across MCP2515 VDD. One across the transceiver VCC.
2	Load capacitors for the crystal	Each crystal leg to ground. About 22 pF when the crystal is marked 18 pF.

2. Window box parts

Buy two radios. Fit one in the window now. Keep the second in the bag until the bike box has passed the fuse test.

Qty	Buy this	Where	Search
1	Espressif ESP32-S3-DevKitC-1. N8R8 is fine.	Espressif's devkit page lists Digi-Key and Mouser. espressif.com/en/products/devkits	ESP32-S3-DevKitC-1
2	Reyax RYLR998 with the built-in antenna. Not RYLR998_M4. That one has a connector and no antenna.	Digi-Key Marketplace 5487-RYLR998-ND showed 5,547 in stock at USD 11.80 on 2 Nov 2026. Datasheet 9 Feb 2026: reyax.com	RYLR998
1	5 V USB power adapter, 1 A, and a short USB cable that fits the devkit.	Any phone charger that is really 5 V. A plastic box for the window sill.	—

3. How the bike box connects

Power is the fused battery lead. The 16-pin plug does not power the box. USB is for programming on the bench. Unplug USB before the sleep test and before the box goes on the bike. Do not feed 12 V into the 5 V pin or the 3V3 pin. 3V3 is an output.

Power path, in order

From	To	Notes
Battery positive post	1 A fuse, within 15 cm	18 AWG red
Fuse output	1N5822 anode (the end without the band)	Band points toward the regulator
1N5822 cathode (banded end)	TPS7A1601 pin 8, IN	Also pin 5, EN, tied to pin 8
That same cathode	P6KE18A banded end	The other end of the TVS goes to ground. A backwards TVS blows the fuse.
TPS7A1601 pin 1, OUT	XIAO BAT+ pad on the back	Meter this at about 3.7 V before the XIAO is connected. 12 V here is a solder bridge.
TPS7A1601 pin 4 and the metal pad	Box ground, and the XIAO GND pad	18 AWG black goes to the battery negative post. A painted frame bolt is a bad ground.
210 kΩ from OUT to pin 2, FB. 100 kΩ from FB to ground.	Sets about 3.70 V	200 kΩ instead of 210 kΩ gives about 3.58 V. That is still acceptable. Wash the flux off.

XIAO nRF52840 pins

Board: Seeed nRF52 Boards, “Seeed XIAO nRF52840”. Board URL to add in Arduino: https://files.seeedstudio.com/arduino/package_seeeduino_boards_index.json. Copy `firmware/bike` to a folder named `bike_link` before you open it. Set `BOX_ID` to the sticker. Leave `radio.h` out until the fuse test passes.

XIAO pin	Connects to	Idle
BAT+	Regulator OUT, about 3.7 V	Always powered from the battery lead
GND	Box ground	
3V3	AO3401A source	Output. On whenever the board is powered.
D0	Divider output, the ADC	Input. 3.3 V zener from D0 to ground, band toward D0.
D1	47 kΩ to the divider-switch 2N3904 base	Low while parked. High only while sampling.
D2	Key-sense 2N3904 collector	Pull-up on. Low means key on.
D3	AO3401A gate	High, or the 100 kΩ pull-up, keeps CAN power off.
D4	MCP2515 pin 16, CS	High while idle. Active low.
D6	Radio TX, later	Leave open on the first build.
D7	Radio RX, later	Leave open on the first build.
D8	MCP2515 pin 13, SCK	SPI clock
D9	MCP2515 pin 15, SO	Data from the CAN controller
D10	MCP2515 pin 14, SI	Data to the CAN controller

D5, the NFC pins, and the on-board battery-read enable (P0.14) stay untouched.

Voltage divider

The sense point is the fuse output, on the battery side of the Schottky, so the diode drop is not in the reading. TO-92, flat face toward you, pins down: emitter, base, collector. That order is the same for the 2N3904 and the 2N3906.

From	To
Fused battery +	2N3906 emitter
2N3906 base	100 kΩ to its own emitter, and 10 kΩ to the 2N3904 collector
2N3906 collector	100 kΩ 1%, then to XIAO D0
XIAO D0	12 kΩ 1% to ground, and the 3.3 V zener to ground
Divider 2N3904 emitter	Ground
Divider 2N3904 base	47 kΩ from D1, and 100 kΩ to ground

D1 high turns the divider on. At 12.6 V the ADC pin should see about 1.35 V. The sketch multiplies by 112 / 12. Trim `BENCH_VOLT_SCALE` until the Bluetooth JSON matches a meter within 0.1 V.

Key sense

From	To
Key-switched 12 V	100 kΩ to the second 2N3904 base
That base	100 kΩ to ground
Collector	XIAO D2
Emitter	Ground

Key on, D2 reads low. The sense wire is OBD pin 16 only when a meter shows about 0 V with the key off and about battery voltage with the key on. If pin 16 stays hot with the key off, leave it open and land this 100 kΩ on a wire that does follow the key, such as the tail lamp.

CAN power switch, AO3401A

SOT-23, text toward you, pins down: pin 1 gate, pin 2 source, pin 3 drain. Confirm the pin-1 mark on the reel you received.

MOSFET pin	Connects to
2, source	XIAO 3V3
3, drain	CAN 3.3 V rail: MCP2515 pin 18 and SN65HVD230 pin 3
1, gate	XIAO D3, and 100 kΩ up to 3V3 so a floating gate keeps the rail off

D3 low turns CAN on. D3 high keeps it off. No 120 Ω resistor anywhere between CAN H and CAN L.

MCP2515-I/P, DIP-18

Pins from the Microchip PDIP datasheet. A DIP socket is fine on the bench. On the bike, solder the chip or strap the socket so vibration cannot walk it out.

Pin	Name	Connects to
1	TXCAN	SN65HVD230 pin 1 (D)
2	RXCAN	SN65HVD230 pin 4 (R)
3	CLKOUT	Open
4, 5, 6	TX0RTS, TX1RTS, TX2RTS	Open
7 and 8	OSC2, OSC1	16 MHz crystal. A load capacitor from each leg to ground.
9	VSS	Ground
10, 11	RX1BF, RX0BF	Open
12	INT	Open. The sketch polls while the key is on.
13	SCK	XIAO D8
14	SI	XIAO D10
15	SO	XIAO D9
16	CS	XIAO D4. Active low.
17	RESET	10 kΩ to the switched 3.3 V rail. Active low.
18	VDD	Switched 3.3 V, and 100 nF to ground

SN65HVD230, SOIC-8

Pins from the TI datasheet. Solder it to the DIP adapter once.

Pin	Name	Connects to
1	D	MCP2515 pin 1
2	GND	Box ground
3	VCC	Switched 3.3 V, and 100 nF to ground
4	R	MCP2515 pin 2
5	Vref	Open
6	CANL	OBD pin 14, twisted pair
7	CANH	OBD pin 6, twisted pair
8	RS	Ground. High-speed while powered. Parked state is power removed.

4. The 16-pin plug

Male SAE J1962 plug into the Triumph's female socket, usually under the seat. Look into the bike's socket, wide edge up. Top row is pins 1–8 left to right. Bottom row is pins 9–16 left to right. Pin 16 is the bottom-right corner. Buzz every pigtail wire to its pin before you trust it. Battery negative disconnected for that check.

1 open	2 open	3 open	4 ground?	5 ground?	6 CAN H	7 open	8 open
9 open	10 open	11 open	12 open	13 open	14 CAN L	15 open	16 sense?

Pin	Wire	Other end
4 or 5	Whichever one has continuity to battery negative	Box ground. If neither does, leave both open. The battery negative wire is the CAN reference.
6	One side of the twisted pair	SN65HVD230 pin 7, CAN H
14	The other side	SN65HVD230 pin 6, CAN L
16	Sense, only if it follows the key	100 kΩ into the key-sense transistor. If it stays hot with the key off, cut it off and insulate it.
Every other pin	None	Cut the pigtail wire off and insulate the end. Pin 7 stays open.

With the bike off and this cable unplugged, resistance between pins 6 and 14 is often about 60 Ω. After the sleeping box is plugged in, that reading should be unchanged. If it falls, unplug the box.

5. How the window box connects

Wall power. No fuse and no sleep budget. Antenna side of the box toward the bikes, in a window at 99 River Street that faces where they actually park. Not a metal electrical room and not a closed cabinet. The RYLR998 has a built-in antenna and no external jack, so the whole box has to be in the window.

Board URL: https://espressif.github.io/arduino-esp32/package_esp32_index.json. Board: ESP32S3 Dev Module. Open `firmware/mailbox/mailbox.ino`. Copy `secrets.example.h` to `secrets.h` in that same folder. Fill in the shop Wi-Fi, the origin `https://service.torontomoto.com` with no path, and the ingest secret. `secrets.h` stays off git. Leave `SOURCE` as `shop_bay` and `PLACE_NAME` as `99 River Street`.

Reyax pin numbers are from the RYLR998 datasheet dated 9 February 2026. The header is the row of five pins. NRST stays open. It already has a pull-up. Holding it low resets the module.

RYLR998 pin	Name	Window box	Bike box, only after the fuse test
1	VDD	ESP32-S3 3V3 pin. Maximum 3.6 V.	XIAO 3V3. Not the 3.7 V battery pad.
2	NRST	Open	Open
3	RXD	Devkit TX, GPIO 17	XIAO D6
4	TXD	Devkit RX, GPIO 18	XIAO D7
5	GND	Devkit GND	XIAO GND

TX and RX are crossed. The sketch sets `AT+CRFOP=0` and `AT+BAND=915000000` in setup, before any send. Confirm with `AT+CRFOP?`. Factory power is 22 dBm. Do not leave it there. On the bike, after each send, the sketch sends `AT+MODE=1` so that module sleeps. The window box stays listening.

USB serial prints every POST body. Until `BIKE_LINK_INGEST_SECRET` is set on the server, type that body into **Record a reading** on the motorcycle page. HTTP 401 prints once and is not retried. The route is `POST /api/bike-link/readings`.

6. Tools

Soldering iron, flux, solder, a magnifier, side cutters, wire strippers, heat-shrink, a multimeter that can read microamps, a current-limited 12 V bench supply, zip ties, dielectric grease, and a drill for the cable gland. Have a motorcycle charger in the bay before the first start. The OBD pigtail arrives with its wires fitted. You do not crimp the 16-pin plug.

7. Assemble the bike box

Build this on the bench. No motorcycle yet. The second radio stays in the bag until both fuse readings are under 50 µA. Leave `radio.h` out of the sketch for this build.

1. Perfboard. Solder the TPS7A1601DGNT first. The metal pad on the bottom goes to ground. Pin 5, EN, ties to pin 8, IN. Pin 3 (PG), pin 6 (NC), and pin 7 (DELAY) stay open. 210 kΩ 1% from OUT (pin 1) to FB (pin 2). 100 kΩ 1% from FB to ground. 10 µF from IN to ground and 10 µF from OUT to ground, close to the chip. Flux, then look for bridges. Do not connect the XIAO.
2. On the red bench lead: 1 A fuse, then the 1N5822 with the band toward the regulator. The banded end of the P6KE18A lands on that same cathode. The other end of the TVS goes to ground. A backwards TVS blows the fuse.
3. Bench supply through the fuse. Meter the regulator output. About 3.7 V. 12 V here is a solder bridge. Stop.
4. Only then solder the regulator output to the XIAO BAT+ pad on the back, and regulator ground to the XIAO GND pad. Do not feed 12 V into the 5 V pin or the 3V3 pin.
5. Copy `firmware/bike` to a folder named `bike_link` and open `bike_link.ino`. Board package URL and board name are in section 3. Set `BOX_ID` to the sticker you will put on the shell. Flash over USB. Unplug USB. The sketch reads fault codes. It does not clear them. Kilometers stay blank until a Triumph in the bay answers a real request.
6. Solder the divider and the key sense, using the tables in section 3. Flat face of each TO-92 toward you, pins down: emitter, base, collector. Zener on D0, band toward D0. D1 stays low while parked.
7. Solder the AO3401A, then the MCP2515-I/P and the SN65HVD230 on its DIP adapter. Confirm the MOSFET pin-1 mark. 16 MHz crystal on MCP2515 pins 7 and 8, a load capacitor from each leg to ground. 100 nF on each CAN chip. No 120 Ω resistor anywhere between CAN H and CAN L. A DIP socket is fine on the bench. On the bike, solder the controller or strap the socket so it cannot walk out.
8. Buzz every pigtail wire to its pin before you trust a color. Cut and insulate every wire except pin 6, pin 14, the ground pin that reaches battery negative, and pin 16 if the meter later shows that it follows the key. Leave CAN H and CAN L open until the bike tests in section 9.
9. Gasketed box, cable gland, drip loop, and the board strapped down. Strain-relieve the wires inside. Dielectric grease on the plug when it goes on the bike. Under the seat, off the chain and out of the wheel spray. Do not pot the box. You still have to open it.

8. Assemble the window box

This can be soldered the same day. It does not go near the bike's fuse. One RYLR998 with the built-in antenna, not the M4. Wall power. No sleep budget.

1. ESP32-S3-DevKitC-1 and the radio in a plastic box. Antenna side of the box toward the bikes, in a window at 99 River Street that faces where they park. Not a metal room and not a closed cabinet. The module has no external antenna jack.
2. Radio pin 1, VDD, to the devkit 3V3 pin. Maximum 3.6 V. Pin 5, GND, to devkit ground. Pin 3, RXD, to GPIO 17. Pin 4, TXD, to GPIO 18. Pin 2, NRST, stays open. TX and RX are crossed.
3. Board URL and board name are in section 5. Open `firmware/mailbox/mailbox.ino`. Copy `secrets.example.h` to `secrets.h` in that folder. Fill in the shop Wi-Fi, `https://service.torontomoto.com` with no path, and the ingest secret. Leave `SOURCE` as `shop_bay` and `PLACE_NAME` as `99 River Street`. `secrets.h` stays off git.
4. Flash, then power it from a 5 V USB adapter, 1 A. The sketch sends `AT+CRFOP=0` and `AT+BAND=915000000` to the radio before it joins Wi-Fi. Confirm with `AT+CRFOP?`. Factory power is 22 dBm. Do not leave it there. The window box stays listening. USB serial prints every POST body. Until the ingest secret is set on the server, type that body into **Record a reading**.
5. The second RYLR998 goes on the bike only after both fuse readings in section 9 are under 50 µA: XIAO 3V3, not the 3.7 V pad, D6 to radio RX, D7 to radio TX, then `radio.h` with `#define BIKE_LINK_RADIO 1`, then reflash. Write the network id and the addresses on the window box.

9. Order of work, and the pass marks

1. Regulator on the bench, through the fuse. No bike. No XIAO. About 3.7 V. Then the battery pads, flash, unplug USB.
2. CAN section. Switch off when D3 is idle. Terminator absent. Measure that it is absent.
3. Whole box from the bench supply, key-sense grounded. 3.3 V on the XIAO 3V3 pin. Asleep, current in series with the fuse under 50 µA. Over that, stop.
4. Bench supply at 12.6 V. The voltage in the Bluetooth JSON matches the meter within 0.1 V.
5. Battery negative off. Fuse at the post. Black wire to battery negative. CAN H and CAN L still open. Reconnect. The bike starts and idles. If it does not, pull the fuse.
6. Key off, asleep, fuse current still under 50 µA.
7. Meter pin 16, key off and key on. Meter pin 4 and pin 5 to battery negative.
8. Battery off again. Connect pin 6, pin 14, and the ground pin that actually reaches battery negative. Start a second time. Resistance between pins 6 and 14 unchanged with the sleeping box plugged in.
9. Stick `TM-BL-` and four digits on the box. On that motorcycle’s page, save the same sticker. Kilometers in the JSON stay blank until a Triumph in the bay answers a real request. Voltage and fault codes are the first reading.
10. Both fuse readings passed. Only then fit the second radio, as in section 8.

The long version of why each part is here, the sleep budget, and the radio limit is `docs/hardware/bike-link-build.md`. The pin tables this sheet was taken from are `docs/hardware/bike-link-wiring.md`. Sketches are `firmware/bike/bike_link.ino` and `firmware/mailbox/mailbox.ino`.